Fiscal Year:	FY 2005	**Task Last Updated:**	FY 03/02/2005
PI Name:	Meiburg, Eckart H. Ph.D.		
Project Title:	The Dynamics of Particulate Deposition and Resuspension Processes at Moderate to High Reynolds Numbers		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Granular flows		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	93106	**Congressional District:**	23
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	98-HEDS-03
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No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
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Flight Program:			
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Key Personnel Changes/Previous PI:	0		
COI Name (Institution):	Maxworthy, Tony (USC) Maxworthy, Tony (USC)		
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	The project consists of an experimental and a numerical component, which will be tuned to each other in order to allow for optimum cross-fertilization. For example, corresponding geometries and parameter ranges will be analyzed in order to allow for comparisons that are as detailed as possible. In addition, each of the components will address parameter values not accessible to the other one, in order to extend the overall range of applicability of the investigation. Furthermore, there are certain quantities for which specific information can be obtained exclusively, or at least more accurately, by only one of the components. Finally, the numerical simulations allow us to include or eliminate certain physical effect, so that information will be obtained that would not easily accessible by experiments only, for example regarding the importance of gravitational effects. In summary, our own past experience has taught us that, if the two components of the investigation are tuned properly, the insight obtained from a joint experimental and numerical		

Task Description:	collaboration can be considerably larger than the sum of individual projects. At the present time, we propose an exploratory investigation into the importance of gravitational effects by means of comparison between a numerical investigation and experiments to be conducted in a conventional, earthbound laboratory. However, we expect this exploratory research to identify specific important issues whose further investigation likely will require a subsequent experimental effort in a microgravity facility. To obtain fundamental insight into the mechanisms that dominate the interaction of a sediment bed with a turbulent boundary layer flowing above it. In particular, the influence of coherent longitudinal vortical structures on the deposition and resuspension processes at the surface of the sediment bed are to be elucidated by means of carefully tuned experimental and computational investigations that account for the coupling of the Navier-Stokes flow with the Darcy Flow inside the sediment bed. It is expected that the detailed understanding to be obtained by comparing the experimental and numerical results will furthermore lead to the formulation of accurate boundary conditions that can be employed for simulating deposition and resuspension phenomena in flows over complex topographies. The transport, deposition, and resuspension of particles embedded in a fluid phase play a dominant role in a wide variety of technological applications and industrial processes. Furthermore, they feature prominently in a host of atmospheric and oceanographic phenomena, such as, for example, the genesis and propagation of powder avalanches and underwater landslides (turbidity currents), the formation and migration of sand dunes, and the evolution of sediment ripples. Of particular interest to NASA?s microgravity program is the lofting and transport of dust on Mars, both by large scale atmospheric winds, as well as by smaller scale air currents generated by technical equipment and machinery, An improved understanding of the physical mechanisms underlying the resuspension and deposition processes in a prerequisite for developing efficient strategies to minimize the negative impact of dust on the operation, performance, and safety of any facility set up on Mars. In addition, the exploitation of natural resources on Mars through mining, the development of in-situ power generation capabilities, and the utilization of Martian soil for habitat construction will lead to the development of industrial processes involving particle laden as well as granular flows. Since the density of the particulate matter usually differs from that of the fluid, gravity significantly affects the dynamics of such multiphase currents. Consequently, the reduced level of gravity on Mars and other celestial bodies is expected to result in scaling laws for the dynamical evolution of particle laden flows that differ significantly from those established in laboratories based on Earth. The derivation of such scaling relationships has been impeded by the lack of sound models for deposition and resuspension processes, due to poor understanding of the underlying physical mechanisms. Besides being of great practical importance, deposition and resuspension phenomena, along with the related topic of granular flows, are also of considerable interest to the physics community on a fundamental level. The formation of well-defined, regular patterns displayed by sand ripples and the related physical instability mechanisms have long been a focus of attention (Kennedy 1969, Engelund and Fredsoe 1982). More recently, topics such as the instabilities in sandpiles, as well as the formation and evolution of avalanches have been investigated within the conceptual framework of ?self-organized criticality? (Kadanoff, Nagel, Wu, and Zhou 1989). The general importance of particle laden flows is also reflected by the recent award of a ?NSF Grand Challenge Project? to the University of Minnesota in this area. The current proposal aims at gaining new insight into the mechanisms underlying sedimentation and suspension processes at moderate to high Reynolds numbers by investigation the interaction between coherent vortical structures of the fluid motion and the sediment bed from a novel point of view. In general terms, our approach will consider the sediment bed as a porous medium of packed particles within which the fluid motion can be described based on Darcy?s law. The flowfield inside the effective porous medium is coupled to the motion of the particle laden fluid above the sediment bed, which is governed by the Navier-Stokes equations. The coupling is enforced by specifying the proper interface conditions for fluid velocity, stresses, and particle transport at the surface of the sediment bed, where particles can settle out by virtue of their settling velocity, or they can be resuspended if the fluidization velocity of the particle bed is exceeded.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	We introduce a computational model for high-resolution simulations of particle-laden gravity currents. The features of the computational model are described in detail, and validation data are discussed. Physical results are presented that focus on the influence of particle entrainment from the underlying bed. As turbulent motions detach particles from the bottom surface, resuspended particles entrained over the entire length of the current are transferred to the current's head, causing it to become denser and potentially accelerating the front of the current. The conditions under which turbidity currents may become self-sustaining through particle entrainment are investigated as a function of slope angle, current and particle size, and particle concentration. The effect of computational domain size and initial aspect ratio of the current on the evolution of the current are also considered. Applications to flows traveling over a surface of varying slope angle, such as turbidity currents spreading down the continental slope, are modeled via a spatially varying gravity vector. Particular attention is given to the resulting particle deposits and erosion patterns.
Bibliography Type:	Description: (Last Updated: 04/27/2010)
Articles in Peer-reviewed Journals	Blanchette, F,; Piche, F., Meiburg, E. "Evaluation of a simplified approach for simulating gravity currents over slopes of varying angles" 8 , Jan-2005